Unit 3 · Session 1 · Tue, Nov 3, 2026

GI & Nutrition Pharmacology — Session 1

4 parts · every fact cited to its slide or textbook page. About 150 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

70 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

20 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

35 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Acid-peptic disease is a **balance**: acid, pepsin, H. pylori and NSAIDs push against mucus, bicarbonate, prostaglandins and epithelial turnover. S5; SG U3-R1

2. Gastrin, histamine and acetylcholine converge on the **proton pump**: **H2 blockers** cut one input, **PPIs** block the final common pathway, **antacids** neutralize acid already made. S8; S12; SG U3-R2

3. PPIs are the most potent but carry long-term risks — **B12 deficiency, hypomagnesemia, fractures, C. difficile** — and omeprazole blunts **clopidogrel** via CYP2C19. S15; SG U3-R3; SG U3-R4

4. **Sucralfate** shields the ulcer base (needs acid); **misoprostol** restores prostaglandins for NSAID-ulcer prevention but is **contraindicated in pregnancy**. S20; SG U3-R5

5. **H. pylori** needs eradication: triple therapy, or **bismuth quadruple** with penicillin allergy, recent macrolide, or clarithromycin resistance; test of cure ≥ 4 weeks later, off the PPI 2 weeks. S23; S25; SG U3-R6

6. Antiemetics block one input to the vomiting center: **5-HT3** (ondansetron), **D2** (metoclopramide — boxed tardive dyskinesia), **NK1** (aprepitant), **H1/M1** (meclizine, scopolamine — motion sickness), plus **CB1** (dronabinol). S30; S36; S38; SG U3-R7; SG U3-R8

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO3.1 Identify major pharmacologic classes used in GI and nutrition conditions and describe their mechanisms of action.

Part 1 · Acid-Peptic Disease & Gastric Acid Secretion
Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors
Part 3 · Mucosal Protectants & H. pylori Eradication
Part 4 · Antiemetics

IO3.2 Describe the indications, contraindications, adverse effects, and clinically significant interactions of GI and nutrition medications.

Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors
Part 3 · Mucosal Protectants & H. pylori Eradication
Part 4 · Antiemetics

IO3.3 Compare agents within and across these classes to distinguish clinically relevant differences in efficacy, safety, and use.

Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors
Part 4 · Antiemetics

IO3.4 Select appropriate pharmacologic therapy for common GI and nutrition conditions based on clinical presentation and patient-specific factors.

Part 3 · Mucosal Protectants & H. pylori Eradication
Part 4 · Antiemetics

IO3.5 Identify appropriate dosing, administration, and monitoring considerations for GI and nutrition medications.

Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors

IO3.6 Explain important medication-use and safety information to patients receiving GI and nutrition pharmacotherapy.

Part 3 · Mucosal Protectants & H. pylori Eradication
Part 4 · Antiemetics

SO1 Identify the major pharmacologic classes used to manage acid-peptic disease and nausea/vomiting, and describe their mechanisms of action.

Part 1 · Acid-Peptic Disease & Gastric Acid Secretion
Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors
Part 3 · Mucosal Protectants & H. pylori Eradication
Part 4 · Antiemetics

SO2 Describe the indications, contraindications, adverse effects, and clinically significant interactions of antisecretory and mucosal-protective agents.

Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors
Part 3 · Mucosal Protectants & H. pylori Eradication

SO3 Compare agents within and across these classes to distinguish clinically relevant differences in efficacy and safety.

Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors
Part 4 · Antiemetics

SO4 Select appropriate therapy for acid-peptic disease and nausea/vomiting based on clinical presentation and patient-specific factors.

Part 3 · Mucosal Protectants & H. pylori Eradication
Part 4 · Antiemetics

SO5 Identify dosing, administration, and monitoring considerations, and explain key medication-safety counseling points to patients.

Part 2 · Antacids, H2 Blockers & Proton Pump Inhibitors
Part 3 · Mucosal Protectants & H. pylori Eradication
Part 4 · Antiemetics

SO6 Preview: GI motility disorders, IBD pharmacotherapy, and Nutrition/Dietary Supplements continue in Session 2 (Thursday, November 5).

⚠ Preview only — taught in Session 2 (L06)

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ABOUT THIS LECTURE

- PA Pharmacology Course · Lecture 3 · Session 1 of 2 — GI & Nutrition Pharmacology. Session 1: Acid-Peptic Disease and Antiemetics (Tuesday, November 3, 8:30–11:00 AM). S1
- Two pathways today: how the stomach makes acid and how we turn it down or protect the mucosa; then how the brain triggers vomiting and which receptor each antiemetic blocks. S1 (notes); S2 (notes)
- Every drug today gets four answers: how it works, who it helps, what can go wrong, and what you tell the patient. S2 (notes)
- Session 2 (Thursday, November 5) picks up with lower-GI motility, IBD, and nutrition/dietary supplements. S2; S2 (notes); S44
- Required text: Katzung's Basic and Clinical Pharmacology, 16th Edition (Todd W. Vanderah, 2024/2025, McGraw-Hill Education; ISBN 1260463303 eBook; AccessMedicine: accessmedicine.mhmedical.com/book.aspx?bookid=3382). Unit 3 — GI & Nutrition: Chapter 62 (Drugs Used in the Treatment of Gastrointestinal Diseases) and Chapter 65 (Dietary Supplements & Herbal Medications). S3; S3 (notes)
- Recommended text (additional support, not a substitute): Pharmacotherapy Principles and Practice (Chisholm-Burns, Schwinghammer, Malone, Kolesar, Lee, 2022, McGraw-Hill Education; Print ISBN 9781260460278, eText ISBN 9781260460285) — supplemental support alongside Katzung's Chapters 62 and 65 for Unit 3. Everything on the slides is testable. S3; S3 (notes)
- Abbreviations used in the drug profiles: DR delayed-release; ER extended-release; ODT orally disintegrating tablet; OTC over the counter; PO by mouth; PR by rectum; IV intravenous; IM intramuscular; BID/TID/QID two/three/four times daily; CrCl creatinine clearance; CKD chronic kidney disease; CNS central nervous system; GERD gastroesophageal reflux disease; NSAID nonsteroidal anti-inflammatory drug; CINV chemotherapy-induced nausea and vomiting; PONV postoperative nausea and vomiting; H2 histamine-2; 5-HT3 serotonin type 3; D2 dopamine type 2; NK1 neurokinin-1; QT QT interval on ECG; CYP/P-gp cytochrome P450/P-glycoprotein; BPH

benign prostatic hyperplasia; mcg microgram; m2 square meter of body surface area. S45; S45 (notes)

- **Original figures in this deck:** Figures 1, 4 and 9 (Gastric Acid Secretion: Where Antisecretory Drugs Act; PPIs vs. H2 Receptor Antagonists: Potency and Duration; The Vomiting Reflex: Where Antiemetics Act) were generated by the course director using the AI tool Claude to illustrate standard, non-copyrighted pharmacologic concepts. They are original diagrams, not reproduced from Katzung's Basic and Clinical Pharmacology or any other copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. https://claude.ai. **Mechanism illustrations** (AI-generated, illustrative, not reproduced from copyrighted sources): H2 blocker action on the parietal cell (Figure 2, slide 14); PPI irreversible pump inhibition (Figure 3, slide 16); sucralfate and misoprostol (Figure 5, slide 21); H. pylori multidrug rationale (Figure 6, slide 24); ondansetron and 5-HT3 blockade (Figure 7, slide 31); receptors in the vomiting pathway (Figure 8, slide 32); metoclopramide D2 blockade and 5-HT4 agonism (Figure 10, slide 37); cannabinoids for chemotherapy-induced nausea (Figure 11, slide 38); cannabinoid CB1 activation (Figure 12, slide 39). Created with an AI image generator under the course director's direction for educational use; the pictures are schematic and not to scale. Figures 2, 3, 5-8 and 10-12 are AI-generated and schematic. S72; S72 (notes)

SLIDE DISCREPANCIES

- **Triple therapy length:** slides say 10–14 days (usual 14); the textbook recommends 14 days and quadruple therapy first-line where clarithromycin resistance is likely. Answer with 14 days. S23; p. 1159
- **Cimetidine's CYP targets:** slides list 1A2, 2C19, 3A4; the textbook lists 1A2, 2C9, 2D6, 3A4. Either way: a broad CYP inhibitor that raises warfarin, phenytoin, theophylline. S14; p. 1157

1 PART 1 · TOPIC

Acid-Peptic Disease & Gastric Acid Secretion

SO1 IO3.1

HIGH-YIELD · KNOW THIS COLD

- Acid-peptic disease reflects an imbalance between mucosal defense and aggressive factors (acid, pepsin, H. pylori, NSAIDs). S44; S5
- H. pylori and NSAIDs cause most PUD. S7

KEY DEFINITIONS

- Acid-peptic disease is **a balance between mucosal defense and acid-driven aggression**. S4
- **Aggressive factors:** gastric acid (HCl), pepsin, H. pylori infection, NSAID use (suppresses protective prostaglandins). S5; SG U3-R1
- **Protective (mucosal defense) factors:** mucus–bicarbonate layer; prostaglandins (stimulate mucus/bicarbonate, maintain mucosal blood flow); epithelial cell turnover. S5; SG U3-R1
- **GERD:** reflux of gastric contents into the esophagus, causing heartburn/regurgitation and, over time, esophageal mucosal injury. S7
- **Peptic ulcer disease (PUD):** a break in the mucosa of the stomach or duodenum; most commonly caused by H. pylori infection or NSAID use. S7
- **Gastritis:** inflammation of the gastric mucosa — acute (e.g., NSAID- or alcohol-related) or chronic (e.g., H. pylori-driven). S7
- TB Over 90% of peptic ulcers are caused by H. pylori infection or NSAIDs; bile also counts as an aggressive factor. p. 1153

HOW IT WORKS

- Disease appears when aggressive factors win or defenses fall: aggressive factors (acid, pepsin, H. pylori, NSAIDs) overwhelm mucus, bicarbonate and prostaglandin-driven blood flow → mucosal injury → GERD, ulcer or gastritis. S5 (notes); S6
- **NSAIDs block COX-1 prostaglandins**, removing mucosal protection — the clearest example of tipping the balance. S5 (notes); S6
- **H. pylori urease makes ammonia**, which injures the mucosa. S6
- **Stress and critical illness raise the risk of stress ulceration.** S6
- **Gastrinoma (Zollinger-Ellison) causes severe acid hypersecretion.** S6
- **Gastrin, histamine, and acetylcholine all converge on the parietal cell proton pump.** H2 blockers cut one input (histamine); PPIs block the last step itself. S8; S8 (notes)
- TB **Physiology of acid secretion:** gastrin acts mainly by releasing histamine from ECL cells; acetylcholine (M3) stimulates the parietal cell directly; somatostatin from antral D cells turns gastrin off. p. 1154

DRUGS NAMED AS EXAMPLES

- Figure 1 is the conceptual spine of the antisecretory section: be able to explain why PPIs are more complete acid suppressors than H2RAs from the diagram alone. S8 (notes)

CLINICAL

- GERD, PUD and gastritis share the same drug toolkit: reduce acid, protect the mucosa, and/or eradicate H. pylori. S7 (notes)
- H. pylori and NSAIDs cause most PUD. S7 (notes)

TRAP

- H2RAs block only the histamine input; gastrin and acetylcholine still drive the pump — PPIs block the final common pathway regardless of which input is active. S8; SG U3-R2

Step by step

How the mucosa gets injured S6

Aggressive factors: acid, pepsin, H. pylori, NSAIDs →

Overwhelm mucus, bicarbonate, prostaglandin blood flow → Mucosal injury →

GERD, ulcer or gastritis

Figure 1 · Gastric acid secretion: where antisecretory drugs act. Original illustrative diagram (not from Katzung): gastrin, histamine, and acetylcholine all converge on the parietal cell proton pump; H2RAs block only the histamine pathway, while PPIs block the final common pathway itself. S8; S8 (notes)

Aggressive vs. protective factors

Aggressive	Protective
Gastric acid (HCl)	Mucus–bicarbonate layer
Pepsin	Prostaglandins (mucus, bicarbonate, mucosal blood flow)
H. pylori (urease → ammonia)	Epithelial cell turnover
NSAIDs (block COX-1 prostaglandins)	—

How to tell them apart. Sources: S5; S6; SG U3-R1

WHY IT HAPPENS

- Why do NSAIDs cause ulcers? They block the COX-1 prostaglandins that make mucus and bicarbonate and keep mucosal blood flow up — they take the defense away. S5 (notes); S6

MNEMONIC (MEMORY AID)

- Three keys, one lock: **gastrin, histamine, ACh** all turn the pump; PPIs change the lock. S8

2 PART 2 · DRUG CLASS

Antacids, H2 Blockers & Proton Pump Inhibitors

SO1 SO2 SO3 SO5 IO3.1 IO3.2 IO3.3 IO3.5

HIGH-YIELD · KNOW THIS COLD

- PPIs block the parietal cell's final common pathway and are the most potent antisecretory agents, but carry real long-term risks (B12 deficiency, hypomagnesemia, fracture, C. difficile). S44; S15
- Antacids: fast, short, don't reduce production; separate other drugs by 2 hours. S12
- Cimetidine: CYP inhibitor + gynecomastia; famotidine: renal dosing, minimal CYP. S13; S14
- Omeprazole + clopidogrel: CYP2C19 inhibition blunts activation. S15; SG U3-R4

CLASS

- Turning down acid production, or shielding the mucosa from it. S11
- **Antacids** neutralize acid already made; **H2 blockers** block the histamine input; **PPIs** block the final common pathway — the most potent acid suppression. S12; S13; S15; S22
- Antacids are the oldest and most "patient-directed" class — a contrast to the receptor- and enzyme-targeted agents. S12 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Omeprazole — PPI prototype (esomeprazole, pantoprazole, lansoprazole, dexlansoprazole). S15
- Famotidine — most commonly used H2 blocker; cimetidine the CYP-inhibiting one. S13
- Calcium carbonate, magnesium hydroxide, aluminum hydroxide — antacids. S12

MECHANISM OF ACTION

- **Antacids — mechanism: directly neutralize existing gastric acid (do not reduce acid production). Acid in stomach → antacid base (Al/Mg hydroxide, calcium carbonate) neutralizes H+ → higher gastric pH** → rapid but short relief. S12

INDICATIONS

- **Antacids:** onset is rapid, but duration is short — best for intermittent, mild symptom relief. S12
- **H2RAs:** mild-to-moderate GERD, maintenance therapy, adjunct for peptic ulcer healing; best at controlling **nocturnal** acid. S13; S14
- **PPIs:** first-line for GERD, PUD healing, and H. pylori regimens. S15

- **H2RAs — mechanism: competitively block histamine at the H2 receptor on parietal cells.** Histamine from ECL cells → H2 receptor → cAMP rises → stimulates the proton pump; famotidine occupies the receptor competitively, so less acid. S13; S14
- **PPIs — mechanism: irreversibly inhibit the H^+/K^+-ATPase proton pump — the final common pathway of acid secretion.** S15; SG U3-R2
- **PPI prodrug activated by acid → binds H^+/K^+ ATPase irreversibly → final step of acid secretion blocked**. S15
- Pathway: enteric-coated prodrug absorbed in the small intestine → reaches the parietal cell canaliculus → activated by acid → binds a cysteine on H^+/K^+-ATPase covalently → **pump irreversibly blocked until new pumps are made**. S16; S16 (notes)
- PPIs achieve substantially greater acid-suppression potency and longer duration of effect than H2RAs (Figure 4: potency 95 vs. 45; duration 90 vs. 50, illustrative). S17; S17 (notes)
- **TB** H2 blockers inhibit 60–70% of 24-hour acid secretion; PPIs inhibit 90–98% because they block the final common pathway. p. 1156; p. 1159
- **TB** Infrequent heartburn (fewer than three times per week): antacids or as-needed H2 blockers; antacids act faster but last 1–2 hours vs. 6–10 hours. p. 1156
- **TB** PPIs heal over 90% of duodenal ulcers within 4 weeks; GERD recurs in over 80% within 6 months of stopping. p. 1159
- **TB** Stress-related bleeding prophylaxis in high-risk ICU patients: an oral PPI if enteral meds are possible, otherwise IV H2 blocker or PPI. p. 1157

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **PPI long-term adverse effects: vitamin B12 deficiency, hypomagnesemia, increased fracture risk, and increased risk of Clostridioides difficile infection** — also pneumonia and acute interstitial nephritis. S15; S16; SG U3-R3
- **Abrupt PPI discontinuation after prolonged use can cause rebound acid hypersecretion.** S15

CONTRAINDICATIONS

- PPIs: contraindicated with rilpivirine (an HIV drug that needs stomach acid). S46; S71
- **TB** Magnesium and aluminum antacids are renally excreted: avoid long-term use in renal insufficiency. p. 1155

- **Aluminum hydroxide — constipating. Magnesium hydroxide — can cause diarrhea; use caution in renal impairment (magnesium accumulation). Calcium carbonate — can cause constipation; risk of milk-alkali syndrome with excessive use.** S12
- **Cimetidine:** antiandrogenic effects (gynecomastia) with prolonged use. S13; S14
- **Famotidine:** confusion risk in older adults with renal impairment. S13 (notes); S14
- **H2RAs: tolerance (tachyphylaxis) can develop with continuous use**, reducing effectiveness over time. S13; S14
- TB PPIs carry an FDA warning for hip, spine and wrist fractures and for life-threatening hypomagnesemia; 2–3-fold more C. difficile infection. p. 1160
- TB Sodium bicarbonate causes belching and metabolic alkalosis; excess calcium carbonate with dairy causes hypercalcemia, renal insufficiency and alkalosis (milk-alkali). p. 1155

DRUG INTERACTIONS

- **Antacids can impair absorption of many other oral drugs — separate dosing by at least 2 hours.** They bind levothyroxine, quinolones, tetracyclines, and iron. S12; S53; S71
- **Omeprazole (a CYP2C19 inhibitor) can reduce clopidogrel activation, blunting its antiplatelet effect.** Omeprazole and esomeprazole both inhibit CYP2C19. S15; S16; SG U3-R4
- **Cimetidine is a potent CYP450 inhibitor (1A2, 2C19, 3A4)** — a callback to Lecture 1: it raises warfarin, phenytoin, and theophylline levels. S13; S13 (notes); S14; S52
- Famotidine has minimal CYP450 interaction. S13

PHARMACOKINETICS

- **PPIs are best absorbed/effective when taken 30–60 minutes before a meal** (timed to peak pump activity) — they work best on actively secreting pumps. S15; S16
- CYP2C19 genotype alters PPI exposure; ultrarapid metabolizers may need higher doses. S15 (notes); S16
- Famotidine needs renal dose adjustment. S13 (notes); S14
- TB PPIs: food cuts bioavailability by about 50%; half-life about 1.5 hours yet acid inhibition lasts up to 24 hours; full effect takes 3–4 days. p. 1158
- TB PPIs need no renal dose reduction; consider reduction in severe liver impairment. p. 1159

- **TB** Antacids should not be given within 2 hours of tetracyclines, fluoroquinolones, itraconazole, and iron. p. 1155
- **TB** With clopidogrel, prefer pantoprazole or rabeprazole (minimal CYP2C19 inhibition). p. 1161
- **TB** Less acid lowers absorption of ketoconazole, itraconazole, digoxin, and atazanavir. p. 1161

MONITORING

- Many patients stay on PPIs far longer than necessary — the long-term adverse-effect list drives deprescribing conversations. S15 (notes)

CLINICAL PEARL / WHY

- Ask: what happens to warfarin or phenytoin levels if cimetidine is added? They rise (CYP inhibition). S13 (notes)
- Name stems: **-prazole** PPIs (irreversibly block H+/K+ ATPase); **-tidine** H2 receptor antagonists (famotidine, ranitidine, cimetidine; competitively block the H2 receptor). S28

Step by step

PPI activation S16

Enteric-coated prodrug absorbed in the small intestine →

Reaches the parietal cell canaliculus → Activated by acid →

Binds cysteine on H+/K+ ATPase covalently →

Pump blocked until new pumps are made

H2 blocker action S14

Histamine from ECL cells → Binds H2 receptor on the parietal cell →

Raises cAMP → Stimulates the proton pump →

Famotidine competes at the receptor → less acid

Figure 2 · H2 blockers: blocking the histamine signal. Schematic (illustrative): ECL cell releases histamine → binds the H2 receptor on the parietal cell → cAMP rises → H+/K+ ATPase pumps move H+ into the gastric lumen. Famotidine blocks the H2 receptor, so less acid is secreted. S14; S14 (notes)

Figure 3 · PPIs: irreversible pump inhibition. Schematic (illustrative): the PPI prodrug (capsule) is absorbed and reaches the parietal cell. In the acidic canaliculus it is activated and binds a cysteine on the H+/K+ ATPase pump (covalent bond). Open pumps move H+ toward the gastric lumen; the blocked pump does not. S16; S16 (notes)

Figure 4 · PPIs vs. H2 receptor antagonists: potency and duration. Original illustrative comparison (illustrative relative values, not from Katzung; not trial data): H2 receptor antagonists (e.g., famotidine) acid-suppression potency 45, duration 50; proton pump inhibitors (e.g., omeprazole) potency 95, duration 90. S17; S17 (notes)

Table 1 · Acid-peptic disease agents at a glance

Class	Example agent(s)	Key teaching point
Antacid	Calcium carbonate, magnesium hydroxide	Neutralizes existing acid; rapid onset, short duration
H2RA	Famotidine, cimetidine	Blocks histamine input only; cimetidine inhibits CYP450
PPI	Omeprazole, pantoprazole	Blocks final common pathway; most potent, long-term AE risks
Mucosal protectant	Sucralfate, misoprostol	Barrier/prostaglandin replacement; misoprostol contraindicated in pregnancy

How to tell them apart. Sources: S22; S22 (notes)

PPI drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Omeprazole (Prilosec)	**Proton pump inhibitor**	DR capsule 10/20/40 mg; OTC 20 mg tablet; oral suspension packets	20-40 mg PO once daily, 30-60 min before breakfast; GERD/ulcer courses 4-8 weeks	GERD, erosive esophagitis, peptic ulcer, H. pylori regimens, NSAID-ulcer prevention, Zollinger-Ellison

Drug	Class	Formulation	Adult dose	Indications
Esomeprazole (Nexium)	**Proton pump inhibitor**	DR capsule 20/40 mg; granule packets; IV vial 20/40 mg	20-40 mg PO daily; IV 20-40 mg over 10-30 min daily when oral route is not possible	GERD, erosive esophagitis, H. pylori, NSAID-ulcer prevention, rebleeding risk reduction
Pantoprazole (Protonix)	**Proton pump inhibitor**	DR tablet 20/40 mg; granule packets; IV vial 40 mg	40 mg PO daily before a meal; IV 40 mg daily for up to 7-10 days	GERD, erosive esophagitis, Zollinger-Ellison; IV when oral therapy is not possible
Lansoprazole (Prevacid)	**Proton pump inhibitor**	DR capsule 15/30 mg; orally disintegrating tablet 15/30 mg	15-30 mg PO daily before a meal; courses 4-8 weeks	GERD, ulcers, H. pylori, NSAID-ulcer healing and prevention
Dexlansoprazole (Dexilant)	**Proton pump inhibitor**	DR capsule 30/60 mg	30 mg daily for non-erosive GERD; 60 mg daily for up to 8 weeks for erosive esophagitis; with or without food	Erosive esophagitis healing and maintenance, non-erosive GERD heartburn

How to tell them apart. Sources: S46; S47; S48; S49; S50

PPI drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Omeprazole (Prilosec)	Headache, diarrhea, abdominal pain, nausea; long-term: low B12 and Mg, fractures, C. difficile	Hypersensitivity to PPIs; rilpivirine co-use	Overdose rare (flushing, tachycardia, confusion); long-term hypomagnesemia; reduces clopidogrel activation (CYP2C19)
Esomeprazole (Nexium)	Headache, diarrhea, nausea; long-term: low B12 and Mg, fractures	Hypersensitivity to PPIs; rilpivirine co-use	Same class toxicities as omeprazole; hypomagnesemia, C. difficile; supportive care for overdose
Pantoprazole (Protonix)	Headache, diarrhea, nausea; long-term: low B12 and Mg, fractures	Hypersensitivity to PPIs; rilpivirine co-use	Class toxicities; fewer CYP drug interactions than omeprazole; supportive care for overdose
Lansoprazole (Prevacid)	Diarrhea, abdominal pain, nausea, headache; long-term: low B12 and Mg	Hypersensitivity to PPIs; rilpivirine co-use	Class toxicities; ODT contains phenylalanine; supportive care for overdose
Dexlansoprazole (Dexilant)	Diarrhea, abdominal pain, nausea, headache; long-term: low B12 and Mg	Hypersensitivity to PPIs; rilpivirine co-use	Class toxicities; supportive care for overdose

How to tell them apart. Sources: S46; S47; S48; S49; S50

H2 blocker and antacid drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Famotidine (Pepcid)	**H2 receptor antagonist**	Tablet 10 mg (OTC), 20/40 mg; IV 10 mg/mL; oral suspension	20 mg BID or 40 mg at bedtime; reduce dose when CrCl is below 60 mL/min	GERD, duodenal and gastric ulcer, hypersecretory states, heartburn prevention
Cimetidine (Tagamet)	**H2 receptor antagonist**	Tablet 200/300/400/800 mg; oral liquid; IV	400 mg BID or 800 mg at bedtime; adjust for renal impairment	Duodenal and gastric ulcer, GERD, hypersecretory states
Calcium carbonate (Tums)	**Antacid and calcium supplement**	Chewable tablet 500-1,000 mg; suspension	500-1,500 mg PO as needed; respect label daily maximum (about 7,000-7,500 mg/day)	Occasional heartburn and indigestion; calcium supplementation
Magnesium hydroxide (Milk of Magnesia)	**Antacid and osmotic laxative**	Suspension 400 mg/5 mL; chewable tablet	Antacid 5-15 mL PO up to four times daily; laxative 30-60 mL once daily	Occasional heartburn, short-term constipation
Aluminum hydroxide (Amphojel)	**Antacid and phosphate binder**	Suspension 320 mg/5 mL; tablet	5-10 mL PO several times daily between meals and at bedtime as labeled	Heartburn, indigestion; hyperphosphatemia in kidney disease

How to tell them apart. Sources: S51; S52; S53; S54; S55

H2 blocker and antacid drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Famotidine (Pepcid)	Headache, dizziness, constipation or diarrhea; confusion in older or renally impaired patients	Hypersensitivity to H2 blockers	Overdose rare: CNS effects, tachycardia, hypotension; accumulates in CKD; tolerance with continued use
Cimetidine (Tagamet)	Headache, dizziness, confusion, gynecomastia, impotence	Hypersensitivity to H2 blockers	Strong CYP inhibitor: raises warfarin, phenytoin, theophylline levels; CNS toxicity in older adults
Calcium carbonate (Tums)	Constipation, belching, gas; hypercalcemia with excess	Hypercalcemia, kidney stones (caution), severe renal impairment	Milk-alkali syndrome (hypercalcemia, alkalosis, AKI); acid rebound; binds levothyroxine, quinolones, tetracyclines, iron

Drug	Adverse effects	Contraindications	Toxicity / overdose
Magnesium hydroxide (Milk of Magnesia)	Diarrhea, abdominal cramping	Severe renal impairment, bowel obstruction	Hypermagnesemia in CKD: hypotension, absent reflexes, respiratory depression, cardiac arrest
Aluminum hydroxide (Amphojel)	Constipation, hypophosphatemia with prolonged use	Bowel obstruction; caution in CKD	Aluminum accumulation in CKD: osteomalacia, encephalopathy, anemia; binds other drugs

How to tell them apart. Sources: S51; S52; S53; S54; S55

WHY IT HAPPENS

- Why before breakfast? PPIs only hit pumps that are actively secreting; a meal switches pumps on, so the drug should be peaking when they turn on. S15; S16
- Why does PPI use cause B12 deficiency? Acid frees B12 from food protein; less acid, less B12 released. SG U3-R3

MNEMONIC (MEMORY AID)

- Antacid bowel effects: **Al**uminum **al**ways constipates; **Mg** = **m**ore **g**oing (diarrhea). S12

3 PART 3 · DRUG CLASS

Mucosal Protectants & H. pylori Eradication

SO1 SO2 SO4 SO5 IO3.1 IO3.2 IO3.4 IO3.6

HIGH-YIELD · KNOW THIS COLD

- H. pylori eradication requires multi-drug regimens (triple or bismuth quadruple therapy), not acid suppression alone. S44; S23
- Misoprostol: PGE1 analog, NSAID-ulcer prevention, never in pregnancy. S20; SG U3-R5
- Sucralfate needs acid; separate from other drugs by 2 hours. S20; S21

CLASS

- Shielding the mucosa (sucralfate, misoprostol, bismuth) and treating the cause (H. pylori eradication). S11; S20; S23
- **H. pylori is the leading identifiable cause of peptic ulcer disease — eradication, not just acid suppression, is the goal.** S23; S24 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Sucralfate — barrier over the ulcer base. S20
- Misoprostol — PGE1 analog for NSAID-ulcer prevention. S20
- Bismuth quadruple and PPI-based triple therapy for H. pylori. S23

MECHANISM OF ACTION

- **Sucralfate:** in acid (pH below 4) it polymerizes, binds the ulcer base and **forms a protective barrier**; requires an acidic environment to activate; minimal systemic absorption. S20; S21; SG U3-R5
- **Misoprostol: a prostaglandin E1 analog** — acts on EP receptors to reduce acid and increase mucus and bicarbonate, **restoring the protective prostaglandin tone that NSAIDs suppress**. S20; S21; SG U3-R5
- **Why H. pylori needs multiple drugs:** it lives in gastric mucus and buffers acid with urease-made ammonia. S24

INDICATIONS

- **Triple therapy:** PPI + amoxicillin + clarithromycin for 10–14 days (usual duration 14 days). S23; S23 (notes)
- **Bismuth quadruple therapy:** PPI (or H2RA) + bismuth subsalicylate + metronidazole + tetracycline — preferred where clarithromycin resistance is a concern, with recent macrolide exposure, or with penicillin allergy. S23; S23 (notes); S24; SG U3-R6
- **Penicillin allergy:** substitute metronidazole for amoxicillin in triple therapy. S23
- Misoprostol is used specifically to prevent NSAID-induced ulcers. S20
- TB Misoprostol cuts NSAID-induced ulcers to less than 3% and ulcer complications by 50%, but dosing 3–4 times daily and diarrhea limit use. p. 1162

- A PPI raises gastric pH so bacteria enter a replicating state and antibiotics are more stable; amoxicillin blocks cell wall synthesis; clarithromycin (50S ribosome) or metronidazole adds a second antibiotic. S24; S24 (notes)
- TB Sucralfate is sucrose complexed to sulfated aluminum hydroxide; less than 3% is absorbed. p. 1161
- TB Bismuth coats ulcers, has direct antimicrobial effects and binds enterotoxins (traveler's diarrhea). p. 1162

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- **Misoprostol: contraindicated in pregnancy — potent uterotonic/abortifacient effect** (boxed warning). Common adverse effects: diarrhea and cramping. S20; S21; S57; SG U3-R5
- Sucralfate: constipation is common; aluminum accumulates in CKD. S20 (notes); S21
- TB Bismuth blackens stool (can mimic GI bleeding) and darkens the tongue; avoid in renal insufficiency. p. 1162

CONTRAINDICATIONS

- Avoid tetracycline in pregnancy and in children. S23 (notes); S24
- Penicillin allergy rules out amoxicillin in H. pylori regimens. S23; S71

DRUG INTERACTIONS

- **Separate sucralfate from other drugs by 2 hours** (give on an empty stomach, separated from antacids/PPIs). S20; S21
- **Clarithromycin inhibits CYP3A4** — must not be combined with simvastatin, lovastatin, or ergotamine. S23 (notes); S24; S71
- **Metronidazole plus alcohol causes a disulfiram-like reaction.** S23 (notes); S24; S71

PHARMACOKINETICS

- Sucralfate has minimal systemic absorption. S20

MONITORING

- **Confirm eradication with testing (urea breath test or stool antigen) after completing therapy** — at least 4 weeks after, with the PPI held for 2 weeks (PPIs cause false negatives). S23; S23 (notes); S24
- Counsel on completing a 10–14 day, 3–4 drug regimen — multi-drug regimens are a classic adherence problem. S23 (notes)
- TB After antibiotics, continue the PPI once daily for a total of 4–6 weeks to complete ulcer healing. p. 1159

CLINICAL PEARL / WHY

- Case: confirmed H. pylori, penicillin allergy, and a macrolide two months ago. **Which eradication regimen is more appropriate for this patient, and why?** S25
- Answer: bismuth quadruple therapy — recent macrolide exposure raises clarithromycin-resistance concern and the allergy rules out standard triple therapy. S25; S25 (notes); SG U3-S
- Misoprostol's pregnancy contraindication links to its other use (labor induction/medical abortion) — mechanism explains the contraindication. S20 (notes)
- Name stems: **-prostol** prostaglandin analogs (misoprostol); **-cillin** penicillins (amoxicillin; blocks cell wall synthesis); **-thromycin** macrolides (clarithromycin; -mycin alone is not reliable, e.g., gentamicin); sucralfate and bismuth subsalicylate share no stem. S28; S28 (notes)

Step by step

H. pylori decision S23; S25

Confirmed H. pylori → Penicillin allergy or macrolide in recent months? →

No → PPI + amoxicillin + clarithromycin, 14 days →

Yes → bismuth quadruple: PPI + bismuth + metronidazole + tetracycline →

Test of cure ≥ 4 weeks later, PPI held 2 weeks

Figure 5 · Sucralfate and misoprostol: strengthening defense. Schematic (illustrative): sucralfate polymerizes in acid and binds the ulcer base, forming a protective barrier. Misoprostol, a PGE1 analog, activates EP receptors on parietal cells to reduce acid and increases mucus and bicarbonate. S21; S21 (notes)

Figure 6 · Why H. pylori needs multiple drugs. Schematic (illustrative): H. pylori lives in the gastric mucus and buffers acid with urease-made ammonia. A PPI raises pH, amoxicillin blocks cell wall synthesis, and clarithromycin or metronidazole adds a second antibiotic; bismuth quadruple therapy (PPI + bismuth + tetracycline + metronidazole) is the alternative. S24; S24 (notes)

H. pylori regimens

Regimen	Drugs	Use when
Triple therapy	PPI + amoxicillin + clarithromycin, 10–14 days	Low clarithromycin resistance, no penicillin allergy
Triple, penicillin-allergic	PPI + metronidazole + clarithromycin	Penicillin allergy (substitute metronidazole for amoxicillin)
Bismuth quadruple	PPI (or H2RA) + bismuth subsalicylate + metronidazole + tetracycline	Clarithromycin resistance, recent macrolide, penicillin allergy
Test of cure	Urea breath test or stool antigen	≥ 4 weeks after therapy, PPI held 2 weeks

How to tell them apart. Sources: S23; S23 (notes); S24; SG U3-R6

Mucosal protectant drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Sucralfate (Carafate)	**Mucosal protectant**	Tablet 1 g; suspension 1 g/10 mL	1 g PO QID, 1 hour before meals and at bedtime, 4-8 weeks (duodenal ulcer); 1 g BID for maintenance	Active duodenal ulcer, ulcer maintenance, stress-ulcer prophylaxis (off-label)
Misoprostol (Cytotec)	**Prostaglandin E1 analog**	Tablet 100/200 mcg	200 mcg PO QID with food and at bedtime during NSAID therapy	Prevention of NSAID-induced gastric ulcers
Bismuth subsalicylate (Pepto-Bismol)	**Antisecretory, antimicrobial, mucosal agent**	Chewable tablet 262 mg; suspension 262 or 525 mg/15 mL	524 mg every 30-60 min as needed, max 8 doses per day; H. pylori quadruple therapy 524 mg QID	Diarrhea, dyspepsia, nausea, H. pylori eradication

How to tell them apart. Sources: S56; S57; S58

Mucosal protectant drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Sucralfate (Carafate)	Constipation, dry mouth, nausea	Hypersensitivity; caution in CKD and dysphagia	Aluminum accumulation in CKD; bezoar risk; binds other drugs (separate by 2 hours); hyperglycemia in diabetes
Misoprostol (Cytotec)	Diarrhea, abdominal cramps, nausea, menstrual-type bleeding	Pregnancy (boxed warning), prostaglandin hypersensitivity	Miscarriage, uterine rupture and birth defects; overdose: fever, tachycardia, hypotension, seizures
Bismuth subsalicylate (Pepto-Bismol)	Black stool and tongue, constipation	Salicylate allergy, bleeding disorders, anticoagulants, children or teens with viral illness (Reye syndrome)	Salicylism: tinnitus, rapid breathing; bismuth neurotoxicity with prolonged high use

How to tell them apart. Sources: S56; S57; S58

H. pylori antibiotic profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Amoxicillin (Amoxil)	**Aminopenicillin**	Capsule/tablet 250/500/875 mg; suspension	H. pylori: 1 g PO BID for 14 days as part of combination therapy	H. pylori (combination), bacterial infections

Drug	Class	Formulation	Adult dose	Indications
Clarithromycin (Biaxin)	**Macrolide antibiotic**	Tablet 250/500 mg; ER tablet 500 mg; suspension	H. pylori: 500 mg PO BID for 14 days in triple therapy	H. pylori (where resistance is low), respiratory infections
Metronidazole (Flagyl)	**Nitroimidazole antimicrobial**	Tablet 250/500 mg; IV; topical/vaginal forms	H. pylori quadruple therapy: 500 mg PO TID-QID for 14 days	H. pylori (bismuth quadruple), anaerobic infections, C. difficile, trichomoniasis
Tetracycline (Sumycin)	**Tetracycline antibiotic**	Capsule 250/500 mg	H. pylori quadruple therapy: 500 mg PO QID for 14 days	H. pylori (bismuth quadruple), acne, rickettsial infections

How to tell them apart. Sources: S59; S60; S61; S62

H. pylori antibiotic profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Amoxicillin (Amoxil)	Diarrhea, nausea, rash; C. difficile infection	Penicillin allergy	Anaphylaxis in allergic patients; overdose causes GI upset; rarely crystalluria
Clarithromycin (Biaxin)	Taste change, nausea, diarrhea; QT prolongation	Macrolide allergy; QT history; co-use with simvastatin, lovastatin, ergotamine	Strong CYP3A4 and P-gp inhibitor; torsades, hepatotoxicity; avoid if prior macrolide exposure
Metronidazole (Flagyl)	Metallic taste, nausea, headache; disulfiram-like reaction with alcohol	Alcohol use during and for 3 days after, recent disulfiram, Cockayne syndrome	Neurotoxicity: neuropathy, encephalopathy, seizures with prolonged or high dosing
Tetracycline (Sumycin)	Nausea, esophagitis, photosensitivity	Pregnancy, children under 8 years, hypersensitivity	Tooth discoloration and bone effects in children; hepatotoxicity at high dose; chelated by dairy, antacids, iron

How to tell them apart. Sources: S59; S60; S61; S62

SLIDE DISCREPANCY

- **H. pylori triple therapy duration**
- Slides: Triple therapy: PPI + amoxicillin + clarithromycin for 10–14 days (usual duration 14 days). S23; S23 (notes)
- Textbook: A 14-day regimen is recommended; quadruple therapy is now first-line where clarithromycin resistance is likely. p. 1159

WHY IT HAPPENS

- Why is misoprostol forbidden in pregnancy? It is a prostaglandin — the same uterotonic effect that makes it a labor-induction drug. S20; S20 (notes)

MNEMONIC (MEMORY AID)

- Allergic to penicillin or recently on a macrolide? **Go Bismuth (quadruple)**. S25; SG U3-R6

4 PART 4 · DRUG CLASS

Antiemetics

SO1 SO3 SO4 SO5 IO3.1 IO3.2 IO3.3 IO3.4 IO3.6

HIGH-YIELD · KNOW THIS COLD

- Antiemetics are organized by the receptor pathway they block — serotonin (5-HT3), dopamine/neurokinin (CTZ), or histamine/acetylcholine (vestibular) — all converging on the vomiting center. S44; S30
- Metoclopramide: boxed tardive dyskinesia; avoid beyond 12 weeks. S36; SG U3-R8
- Ondansetron: QT prolongation; give before chemo. S31

CLASS

- **Four receptor systems feed one final common reflex.** S29
- Four inputs converge on the **vomiting center in the medulla**: **gut and vagal (5-HT3)**, the **chemoreceptor trigger zone (D2, 5-HT3, NK1)**, the **vestibular (H1, M1)** system, and **higher cortical (anxiety, anticipatory)** input — benzodiazepines and cannabinoids. S32; S32 (notes)
- **Motion sickness is vestibular** — H1/M1 blockers (meclizine, scopolamine) fit it. S33 (notes); SG U3-R7
- Matching each drug to its input pathway (Figure 9) is the single most testable concept in this section. S33 (notes)

PROTOTYPE + OTHER DRUGS NAMED

- Ondansetron — 5-HT3 antagonist. S30
- Metoclopramide, prochlorperazine — D2 antagonists. S30
- Aprepitant — NK1 antagonist. S30
- Meclizine, scopolamine — H1/M1 antagonists. S30
- Dronabinol, nabilone — cannabinoids (CB1). S38

MECHANISM OF ACTION

- **5-HT3 antagonists** block serotonin at gut vagal afferents/CTZ. S30
- Chemotherapy or gut irritation damages enterochromaffin cells → serotonin binds 5-HT3 on vagal afferents and the CTZ → signal to the vomiting center; **ondansetron blocks 5-HT3, so no signal reaches the vomiting center**. S30; S31; S31 (notes)

INDICATIONS

- Ondansetron: give before chemotherapy; oral ODT and IV forms are available. S31
- **Metoclopramide clinical uses: nausea/vomiting and diabetic gastroparesis.** S36

- **D2 antagonists** block dopamine at the CTZ (metoclopramide, prochlorperazine). S30
- **NK1 antagonists** block substance P/neurokinin at the CTZ (aprepitant). S30
- **H1/M1 antagonists** block histamine/acetylcholine from the vestibular system (meclizine, scopolamine). S30
- **Metoclopramide — mechanism: D2 antagonism (antiemetic effect) plus enhancement of upper-GI motility (prokinetic effect).** D2 blockade in the CTZ → less nausea; D2 blockade plus **5-HT4 agonism** in the gut → more acetylcholine → faster gastric emptying. S36; S37; S37 (notes)
- Dronabinol (CB1 agonist) crosses the blood-brain barrier and activates CB1 receptors in the vomiting center, CTZ and hypothalamus, reducing neurotransmitter release → less nausea, more appetite. S38; S39; S39 (notes)
- Mechanism is distinct from the four pathways (5-HT3, D2, NK1, H1/M1). S38; S39; S39 (notes); SG U3-M
- TB **Pathophysiology of vomiting:** the vomiting center sits in the lateral medullary reticular formation; the CTZ (area postrema) lies outside the blood-brain barrier. p. 1169

- **Dronabinol — a synthetic cannabinoid (CB1 receptor agonist) used for refractory chemotherapy-induced nausea/vomiting and appetite stimulation.** Typically reserved for patients who do not respond adequately to standard antiemetics. S38; S38 (notes)
- **Chemotherapy:** combine classes with different targets — a 5-HT3 antagonist, an NK1 antagonist, and dexamethasone; avoid duplicating the same mechanism. S32; S32 (notes)
- **Match drug to cause:** gut/vagal triggers respond to 5-HT3 antagonists; the CTZ to D2 or NK1 antagonists. S32
- TB 5-HT3 antagonists control vagal and chemotherapy emesis but poorly control motion sickness; palonosetron also prevents delayed emesis. p. 1170
- TB **Pharmacokinetics & pharmacodynamics (5-HT3):** first-generation agents have half-lives of 4–9 hours; palonosetron 40 hours; no dose cut in renal insufficiency. p. 1170
- TB NK1 antagonist + 5-HT3 antagonist + dexamethasone prevents acute emesis in 80–90% of highly emetogenic chemotherapy patients. p. 1171
- TB Benzodiazepines (lorazepam, diazepam) prevent anticipatory vomiting before chemotherapy. p. 1172

ADVERSE EFFECTS (BOXED WARNINGS FIRST)

- Metoclopramide — **boxed warning: risk of tardive dyskinesia, particularly with prolonged use or high doses — limit duration of therapy.** Avoid beyond 12 weeks. S36; S37; SG U3-R8
- **Other extrapyramidal symptoms (acute dystonia, akathisia) can also occur**; acute dystonia is treated with diphenhydramine or benztropine. S36; S37; S71

CONTRAINDICATIONS

- **Metoclopramide: contraindicated in GI obstruction; avoid in Parkinson disease.** S36 (notes); S37

- **Ondansetron: QT prolongation** — caution with other QT-prolonging drugs and electrolyte problems. **Common effects: constipation, headache; serotonin syndrome risk with serotonergic drugs.** S31
- **Cannabinoids: sedation, dizziness, mood/perception changes** (euphoria or dysphoria), tachycardia, orthostatic hypotension — counsel patients accordingly. S38; S38 (notes); S39
- TB Metoclopramide: restlessness, drowsiness and agitation in 10–20%; extrapyramidal effects in 25% at high doses; prolactin rises (galactorrhea, gynecomastia). p. 1164
- TB Droperidol prolongs the QT interval (rare fatal torsades) — only if alternatives fail. p. 1172

DRUG INTERACTIONS

- Ondansetron: serotonin syndrome risk with other serotonergic drugs; QT risk with other QT-prolonging drugs. S31
- Cannabinoids: **additive CNS depression with sedatives and alcohol**. S38 (notes); S39
- TB Aprepitant is a CYP3A4 substrate/inhibitor: lowers INR with warfarin; dexamethasone dose is often reduced with it. p. 1171

PHARMACOKINETICS

- **CYP2D6 ultrarapid metabolizers may respond less well to ondansetron.** S31
- **Metoclopramide: renal dosing needed; CYP2D6 poor metabolizers need lower doses** (higher exposure). S36 (notes); S37

MONITORING

- **Dronabinol is Schedule III; nabilone is Schedule II.** Use with caution in older adults. S38 (notes); S39
- Metoclopramide duration matters — many students remember only that it helps nausea. S36 (notes)

CLINICAL PEARL / WHY

- Case: first-trimester nausea and vomiting not helped by dietary changes. **Which antiemetic class has the most established safety profile for use in this patient?** S40
- Pyridoxine (vitamin B6), often with doxylamine, is typically first-line; ondansetron is commonly used too, though first-trimester data continue to evolve — weigh maternal benefit against fetal risk and consult current guidance. S40; S40 (notes); S71
- Pyridoxine is a vitamin — a preview of Session 2's vitamins and supplements. S40 (notes)
- Cannabinoid pharmacology is an expanding area in the newest Katzung edition. S38 (notes)

Step by step

Pick the antiemetic by the trigger S32; S38

Chemotherapy / gut irritation → 5-HT3 (+ NK1 + dexamethasone) →

CTZ / drugs → D2 or NK1 → Motion sickness → H1/M1 (meclizine, scopolamine)

→ Anticipatory → benzodiazepine → Refractory CINV → dronabinol

Figure 7 · Ondansetron: blocking serotonin at 5-HT3. Schematic (illustrative): chemotherapy or gut irritation damages enterochromaffin cells, which release serotonin. Serotonin normally binds 5-HT3 receptors on vagal afferents and the chemoreceptor trigger zone, signaling the vomiting center in the medulla. Ondansetron blocks the 5-HT3 receptor, so no signal reaches the vomiting center. S31; S31 (notes)

Figure 8 · Receptors in the vomiting pathway. Schematic (illustrative): four inputs converge on the vomiting center in the medulla — the GI tract via the vagus nerve (5-HT3), the chemoreceptor trigger zone (D2, 5-HT3, NK1), the vestibular system (H1, M1) and the cerebral cortex (anxiety, anticipation). The vomiting center then triggers emesis. S32; S32 (notes)

Figure 9 · The vomiting reflex: where antiemetics act. Original illustrative diagram (not from Katzung): serotonin, dopamine/neurokinin, and histamine/acetylcholine signals converge on the vomiting center; ondansetron, metoclopramide/prochlorperazine/aprepitant, and meclizine/scopolamine each block a different input. S33; S33 (notes)

Figure 10 · Metoclopramide: D2 blockade and 5-HT4 agonism. Schematic (illustrative): in the chemoreceptor trigger zone, metoclopramide blocks D2 receptors, reducing nausea. In the gut, D2 blockade and 5-HT4 agonism increase acetylcholine release from enteric neurons, strengthening contraction and speeding gastric emptying. S37; S37 (notes)

Figure 11 · Cannabinoids for chemotherapy-induced nausea. Schematic (illustrative): oral dronabinol is absorbed, crosses the blood-brain barrier and activates CB1 receptors in the vomiting center and hypothalamus, reducing nausea and stimulating appetite. S38

Figure 12 · Cannabinoids: CB1 activation in the brain. Schematic (illustrative): dronabinol or nabilone binds CB1 receptors on presynaptic neurons in the vomiting center and chemoreceptor trigger zone. This reduces neurotransmitter release across the synaptic cleft, so less nausea and vomiting, and appetite increases. S39; S39 (notes)

Antiemetic classes by target

Target	Where	Examples	Watch for
5-HT3	Gut vagal afferents, CTZ	Ondansetron, granisetron	QT, constipation, headache, serotonin syndrome
D2	CTZ	Metoclopramide, prochlorperazine	Extrapyramidal effects; tardive dyskinesia (metoclopramide boxed)
NK1	CTZ	Aprepitant, fosaprepitant	CYP3A4 interactions
H1 / M1	Vestibular	Meclizine, scopolamine	Anticholinergic effects; best for motion sickness
CB1	Vomiting center, CTZ, hypothalamus	Dronabinol, nabilone	Psychoactive; Schedule III / II

How to tell them apart. Sources: S30; S32; S38; SG U3-R7; SG U3-M

Name clues: antiemetics

Stem	Drug class	Examples	Key point
-setron	5-HT3 antagonists	Ondansetron, granisetron	Block serotonin at 5-HT3 receptors
-pitant	NK1 antagonists	Aprepitant, fosaprepitant	Block substance P at NK1 receptors
-azine	Phenothiazines	Prochlorperazine, promethazine	Block D2 receptors and other receptors

Stem	Drug class	Examples	Key point
-izine	H1 antihistamines	Meclizine	Block H1 receptors in the vestibular system
-pramide	Benzamide D2 antagonists	Metoclopramide	Block D2; 5-HT4 agonism adds prokinetic effect
-sone	Corticosteroid	Dexamethasone	Adds to other antiemetics in chemotherapy
None shared	Cannabinoids	Nabilone, dronabinol	Activate CB1 receptors
None shared	Anticholinergic	Scopolamine	Block muscarinic receptors

How to tell them apart. Sources: S43; S43 (notes)

Antiemetic drug profiles: formulation, dose, indications

Drug	Class	Formulation	Adult dose	Indications
Ondansetron (Zofran)	**5-HT3 receptor antagonist**	Tablet 4/8 mg; ODT 4/8 mg; oral film; IV/IM 2 mg/mL	4-8 mg PO/ODT every 8 hours; IV 4 mg; chemotherapy regimens use 8-24 mg pre-dose; single IV dose max 16 mg	Chemotherapy, postoperative and radiation nausea; gastroenteritis and pregnancy nausea (off-label)
Metoclopramide (Reglan)	**Dopamine D2 antagonist, prokinetic**	Tablet 5/10 mg; ODT; oral solution; IV/IM 5 mg/mL	Gastroparesis: 10 mg PO 30 min before meals and at bedtime (max 40 mg/day); limit to 12 weeks; lower dose in renal impairment	Diabetic gastroparesis, GERD (short-term), chemotherapy and postoperative nausea
Aprepitant (Emend)	**NK1 receptor antagonist**	Capsule 40/80/125 mg; IV fosaprepitant 150 mg	CINV: 125 mg PO day 1, then 80 mg days 2-3, with a 5-HT3 antagonist and dexamethasone; PONV: 40 mg pre-op	Prevention of chemotherapy and postoperative nausea and vomiting
Meclizine (Antivert)	**H1 antihistamine, anticholinergic**	Tablet 12.5/25/50 mg; chewable tablet	Motion sickness: 25-50 mg 1 hour before travel, repeat every 24 hours; vertigo: 25-100 mg/day divided	Motion sickness, vertigo
Scopolamine (Transderm Scop)	**Muscarinic antagonist**	Transdermal patch 1.5 mg (about 1 mg over 3 days)	1 patch behind the ear at least 4 hours before travel; replace every 72 hours	Motion sickness, postoperative nausea

Drug	Class	Formulation	Adult dose	Indications
Prochlorperazine (Compazine)	**Phenothiazine D2 antagonist**	Tablet 5/10 mg; suppository 25 mg; IV/IM 5 mg/mL	5-10 mg PO every 6-8 hours (max 40 mg/day); 25 mg PR every 12 hours; 5-10 mg IV/IM	Severe nausea and vomiting, psychosis
Dronabinol (Marinol) and nabilone (Cesamet)	**Cannabinoids**	Dronabinol capsule 2.5/5/10 mg, solution; nabilone capsule 1 mg	Dronabinol 5 mg/m2 1-3 hours before chemotherapy, then every 2-4 hours; appetite: 2.5 mg BID; nabilone 1-2 mg BID	Refractory chemotherapy nausea; anorexia in AIDS (dronabinol)
Dexamethasone (Decadron)	**Corticosteroid**	Tablet 0.5-6 mg; oral solution; IV 4 mg/mL	CINV: 8-12 mg PO/IV on day 1, then 8 mg daily on days 2-4 depending on regimen	Antiemetic adjunct in chemotherapy; inflammation and allergy

How to tell them apart. Sources: S63; S64; S65; S66; S67; S68; S69; S70

Antiemetic drug profiles: adverse effects, contraindications, toxicity

Drug	Adverse effects	Contraindications	Toxicity / overdose
Ondansetron (Zofran)	Headache, constipation, dizziness; QT prolongation	Congenital long QT syndrome; apomorphine co-use	QT prolongation and torsades; serotonin syndrome (rare); transient vision loss
Metoclopramide (Reglan)	Drowsiness, restlessness, diarrhea; depression	GI obstruction, perforation or bleeding; pheochromocytoma; seizure disorder; history of tardive dyskinesia	Boxed warning: tardive dyskinesia (may be irreversible); acute dystonia (diphenhydramine or benztropine); neuroleptic malignant syndrome
Aprepitant (Emend)	Fatigue, constipation, hiccups, raised liver enzymes	Pimozide co-use; hypersensitivity	CYP3A4 substrate and moderate inhibitor: lowers warfarin and hormonal contraceptive effect; overdose drowsiness, headache
Meclizine (Antivert)	Drowsiness, dry mouth, blurred vision, urinary retention	Hypersensitivity; caution in glaucoma, BPH, asthma, older adults	Anticholinergic toxicity: agitation, hallucinations, tachycardia, seizures; additive sedation
Scopolamine (Transderm Scop)	Dry mouth, drowsiness, blurred vision, dizziness	Narrow-angle glaucoma; hypersensitivity	Anticholinergic toxicity; withdrawal nausea and dizziness; wash hands to avoid dilated pupils

Drug	Adverse effects	Contraindications	Toxicity / overdose
Prochlorperazine (Compazine)	Sedation, hypotension, dry mouth, extrapyramidal symptoms	Children under 2 years, coma, dementia-related psychosis (boxed warning)	Acute dystonia, tardive dyskinesia, neuroleptic malignant syndrome, QT prolongation, sedation
Dronabinol (Marinol) and nabilone (Cesamet)	Dizziness, euphoria, drowsiness, tachycardia	Hypersensitivity; sesame oil allergy (dronabinol capsule)	Psychoactive: anxiety, paranoia, tachycardia, hypotension; additive CNS depression; misuse potential
Dexamethasone (Decadron)	Insomnia, hyperglycemia, mood change, fluid retention	Systemic fungal infection; hypersensitivity	Adrenal suppression with prolonged use; infection, hyperglycemia, psychosis; taper after long courses

How to tell them apart. Sources: S63; S64; S65; S66; S67; S68; S69; S70

Supporting drugs at a glance (listed for context; not profiled in detail)

Drug(s)	Why it matters
Diphenhydramine and benztropine	Rescue for acute dystonia from metoclopramide or prochlorperazine
Simvastatin, lovastatin, ergotamine	CYP3A4 substrates that must not be combined with clarithromycin
Doxylamine plus pyridoxine (B6)	Nausea and vomiting of pregnancy
Disulfiram	Alcohol-aversion drug; metronidazole can cause a similar reaction with alcohol
Clopidogrel	Prodrug activated by CYP2C19; omeprazole and esomeprazole can blunt its effect
Levothyroxine and quinolones	Absorption falls when taken with antacids; separate doses
Rilpivirine	HIV drug that needs stomach acid, so PPIs are contraindicated
Penicillin	Allergy rules out amoxicillin in H. pylori regimens

How to tell them apart. Sources: S71; S71 (notes)

Prokinetic agents in the reading (textbook)

Drug	How it speeds the gut	Point
Metoclopramide, domperidone	D2 blockade removes dopamine’s brake on cholinergic stimulation	Gastroparesis first-line; domperidone barely enters the brain

Drug	How it speeds the gut	Point
Cholinomimetic agents: bethanechol, neostigmine	M3 agonist; acetylcholinesterase inhibitor	Bethanechol seldom used; IV neostigmine for acute colonic pseudo-obstruction
Macrolides (erythromycin)	Motilin receptor agonist	Gastroparesis trial; tolerance develops rapidly

How to tell them apart. Sources: p. 1163; p. 1164

WHY IT HAPPENS

- Why does ondansetron fail for motion sickness? Motion sickness comes through the vestibular system (H1/M1), not the gut's serotonin pathway. S32; S33 (notes)
- Why combine ondansetron, aprepitant and dexamethasone? Each hits a different target; stacking different mechanisms beats doubling one. S32

MNEMONIC (MEMORY AID)

- **SeT-D-NK-HM:** ondan**set**ron 5-HT3 · metoclopramide **D**2 · apre**pitant** NK1 · meclizine/scopolamine **H**1/**M**1 — plus dronabinol CB1. S30; SG U3-M

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Aggressive factors	Acid, pepsin, H. pylori, NSAIDs S5; SG U3-R1
Protective factors	Mucus–bicarbonate, prostaglandins, epithelial turnover S5; SG U3-R1
Antacids	Neutralize acid; fast, short; separate drugs by 2 h S12
Al vs. Mg vs. Ca antacid	Al constipates · Mg diarrhea (caution CKD) · Ca constipates, milk-alkali S12
Famotidine	H2 block; renal dosing; confusion in older adults; minimal CYP S13; S14
Cimetidine	CYP inhibitor (1A2, 2C19, 3A4); gynecomastia S13; S14
H2 blocker strength	Nocturnal acid; tachyphylaxis S14
PPI mechanism	Irreversible H+/K+-ATPase block (final common pathway) S15; SG U3-R2
PPI timing	30–60 min before the first meal S15; S16
Long-term PPI risks	B12, Mg, fractures, C. difficile, pneumonia, interstitial nephritis, rebound acid S15; S16; SG U3-R3
Omeprazole + clopidogrel	CYP2C19 inhibition blunts activation S15; SG U3-R4
Sucralfate	Barrier in acid; 2 h from other drugs; aluminum in CKD S20; S21
Misoprostol	PGE1 analog; NSAID ulcers; NOT in pregnancy; diarrhea S20; SG U3-R5
Triple therapy	PPI + amoxicillin + clarithromycin, 14 days S23
Bismuth quadruple	PPI + bismuth + metronidazole + tetracycline S23; SG U3-R6
Penicillin allergy (triple)	Metronidazole replaces amoxicillin S23
Test of cure	≥ 4 weeks after; PPI held 2 weeks S23; S24
Clarithromycin / metronidazole	CYP3A4 (statins) / disulfiram-like with alcohol S24
Antiemetic targets	5-HT3 ondansetron · D2 metoclopramide · NK1 aprepitant · H1/M1 meclizine, scopolamine · CB1 dronabinol S30; SG U3-M
Motion sickness	Vestibular → H1/M1 S33; SG U3-R7
Ondansetron	QT; constipation; serotonin syndrome; give before chemo S31

Prompt	Answer
Metoclopramide	Boxed tardive dyskinesia; ≤ 12 weeks; no obstruction/Parkinson S36; S37; SG U3-R8
Chemo combo	5-HT3 + NK1 + dexamethasone S32
Pregnancy nausea	Pyridoxine (B6) ± doxylamine first-line S40
Cannabinoids	CB1; dronabinol C-III, nabilone C-II S38; S39
Stems	-prazole PPI · -tidine H2 · -prostol PG · -setron 5-HT3 · -pitant NK1 · -izine H1 S28; S43
H2 blocker vs. PPI suppression	60–70% vs. 90–98% of 24-hour acid p. 1156; p. 1159
PPI timing	30–60 min before breakfast; full effect in 3–4 days p. 1158
PPI with clopidogrel	Prefer pantoprazole or rabeprazole p. 1161
Antacid spacing	2 hours from tetracyclines, fluoroquinolones, itraconazole, iron p. 1155
Bismuth	Black stool and tongue; avoid in renal insufficiency p. 1162
CINV triple combo	NK1 + 5-HT3 + dexamethasone: 80–90% prevention p. 1171

Self-test

First, the lecture's own 5 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S9-10

Based on Figure 1, why do PPIs achieve more complete acid suppression than H2 receptor antagonists?

- **A** PPIs block the final common pathway (the proton pump itself), so they work regardless of which upstream signal is driving acid secretion
- **B** PPIs block only the gastrin pathway, which is the strongest stimulus
- **C** H2RAs block the proton pump directly, while PPIs act upstream
- **D** There is no mechanistic difference between the two classes

ANSWER: A

H2RAs block only one of three inputs to the parietal cell (histamine), so gastrin and acetylcholine can still drive acid secretion. PPIs irreversibly inhibit the H^+/K^+-ATPase proton pump — the single convergence point for all three signals — producing more complete and longer-lasting acid suppression. S9-10

Lecture check 2 · S18-19

A patient taking clopidogrel after a cardiac stent is started on omeprazole for GERD. What is the primary concern with this combination?

- **A** Omeprazole may reduce CYP2C19-mediated activation of clopidogrel, blunting its antiplatelet effect
- **B** Omeprazole increases clopidogrel absorption, raising bleeding risk
- **C** There is no clinically relevant interaction between these two drugs
- **D** Omeprazole should never be combined with any cardiac medication

ANSWER: A

Clopidogrel is a prodrug that requires CYP2C19 activation. Omeprazole inhibits CYP2C19, which can reduce clopidogrel's active metabolite and blunt its antiplatelet effect — a clinically important interaction, especially soon after stent placement. S18-19

Lecture check 3 · S26-27

A patient is started on misoprostol for NSAID-induced ulcer prevention. What is the single most important question to ask before prescribing?

- **A** Is the patient taking any antacids?
- **B** Is there any possibility of pregnancy?
- **C** Does the patient have a penicillin allergy?
- **D** Is the patient on a PPI already?

ANSWER: B

Misoprostol is a potent uterotonic and is contraindicated in pregnancy due to abortifacient risk — pregnancy status must be ruled out (and reliable contraception confirmed) before starting therapy. S26-27

Lecture check 4 · S34-35

A patient with motion sickness needs prophylaxis before a cruise. Based on Figure 9, which drug class targets the pathway most relevant to this cause of nausea?

- **A** H1 / M1 antagonists (e.g., meclizine, scopolamine) — targeting the vestibular system
- **B** 5-HT3 antagonists (e.g., ondansetron) — targeting gut/chemo input
- **C** NK1 antagonists (e.g., aprepitant) — targeting the CTZ
- **D** D2 antagonists (e.g., metoclopramide) — targeting the CTZ

ANSWER: A

Motion sickness is driven primarily by vestibular system input, which signals through histamine (H1) and muscarinic (M1) receptors — exactly the pathway meclizine and scopolamine block in Figure 9. The other classes target gut/chemo (5-HT3) or CTZ (D2, NK1) inputs, which are not the primary driver here. S34-35

Lecture check 5 · S41-42

A patient has been on metoclopramide for gastroparesis for several months. What is the primary long-term safety concern that should limit duration of therapy?

- **A** Hepatotoxicity
- **B** Tardive dyskinesia
- **C** QT prolongation
- **D** Nephrotoxicity

ANSWER: B

Metoclopramide carries a boxed warning for tardive dyskinesia, with risk increasing with duration of use and cumulative dose — therapy should generally be limited to the shortest effective duration (typically no more than 12 weeks). S41-42

Original questions

Q1 · Mechanism · Acid-Peptic Disease & Gastric Acid Secretion

Which set lists only protective (mucosal defense) factors?

- **A** Pepsin, gastrin, histamine, acetylcholine
- **B** Acid, pepsin, prostaglandins, epithelial turnover
- **C** NSAIDs, urease, ammonia, bicarbonate
- **D** H. pylori, mucus, bicarbonate, NSAIDs
- **E** Mucus–bicarbonate layer, prostaglandins, epithelial turnover

ANSWER: E

Protective factors are the mucus–bicarbonate layer, prostaglandins, and epithelial cell turnover (S5; SG U3-R1). S5; SG U3-R1

- **A.** These drive acid secretion or digestion (S5; S8).
- **B.** Acid and pepsin are aggressive factors (S5).
- **C.** NSAIDs, urease and ammonia injure the mucosa (S6).
- **D.** H. pylori and NSAIDs are aggressive (S5).

Q2 · Mechanism · Acid-Peptic Disease & Gastric Acid Secretion

In Zollinger-Ellison syndrome (gastrinoma), what drives the ulcer disease?

- **A** Urease-produced ammonia injuring the mucosa
- **B** Severe acid hypersecretion from excess gastrin
- **C** Stress ulceration from critical illness
- **D** Loss of COX-1 prostaglandins from NSAID use
- **E** Reflux of gastric contents into the esophagus

ANSWER: B

Gastrinoma (Zollinger-Ellison) causes severe acid hypersecretion (S6). S6

- **A.** That is H. pylori (S6).
- **C.** A separate risk (S6).
- **D.** That is the NSAID mechanism (S6).
- **E.** That defines GERD (S7).

Q3 · Mechanism · Antacids, H2 Blockers & Proton Pump Inhibitors

Which statement describes how PPIs work?

- **A** Bases that neutralize acid already in the stomach
- **B** PGE1 analogs that raise mucus and bicarbonate
- **C** Polymers that coat the ulcer base in acid
- **D** Acid-activated prodrugs that covalently block the proton pump
- **E** Competitive blockers of histamine at the H2 receptor

ANSWER: D

PPIs are enteric-coated prodrugs activated by acid in the canaliculus that bind the H^+/K^+-ATPase covalently until new pumps are made (S15–S16). S15; S16

- **A.** That is antacids (S12).
- **B.** That is misoprostol (S20).
- **C.** That is sucralfate (S20).
- **E.** That is H2RAs (S13).

Q4 · Clinical vignette · Antacids, H2 Blockers & Proton Pump Inhibitors

A patient on long-term omeprazole asks what to monitor. Which set reflects the class-wide long-term risks?

- **A** Thyroid function and prolactin
- **B** INR and bleeding time only
- **C** Hepatic transaminases only
- **D** Serum potassium and digoxin level only
- **E** B12, magnesium, fracture risk, C. difficile

ANSWER: E

Long-term PPI risks: B12 deficiency, hypomagnesemia, fractures, C. difficile (plus pneumonia, interstitial nephritis) (S15–S16; SG U3-R3). S15; S16; SG U3-R3

- **A.** Prolactin/gynecomastia relates to cimetidine (S13).
- **B.** Not a class-wide PPI risk (S15).
- **C.** Not the taught list (S15).
- **D.** Not the taught list (S15).

Q5 · Clinical vignette · Antacids, H2 Blockers & Proton Pump Inhibitors

A patient taking levothyroxine starts calcium carbonate for heartburn. What counseling is needed?

- **A** Switch to cimetidine to avoid any interaction
- **B** Take them together to improve absorption
- **C** Separate the doses by at least 2 hours
- **D** No counseling — antacids act only locally
- **E** Double the levothyroxine dose permanently

ANSWER: C

Antacids impair absorption of many oral drugs, including levothyroxine and quinolones — separate by at least 2 hours (S12; S53; S71). S12; S53

- **A.** Cimetidine inhibits CYP450 (S13).
- **B.** Antacids reduce absorption (S12).
- **D.** They bind other drugs (S12).
- **E.** Separation, not dose change, is taught (S12).

Q6 · Clinical vignette · Antacids, H2 Blockers & Proton Pump Inhibitors

An 82-year-old with CKD on famotidine becomes confused. What is the likely explanation?

- **A** Rebound acid hypersecretion causing delirium
- **B** Hypomagnesemia from long-term H2 blocker use
- **C** Milk-alkali syndrome from the H2 blocker
- **D** Cimetidine-like CYP inhibition raising other drugs
- **E** Famotidine accumulated without renal dose adjustment

ANSWER: E

Famotidine needs renal dose adjustment and can cause confusion in older adults with renal impairment (S13 notes; S14; S51). S14; S51

- **A.** Rebound acid follows stopping PPIs (S15).
- **B.** Hypomagnesemia is a PPI risk (S15).
- **C.** That is excess calcium carbonate (S12).
- **D.** Famotidine has minimal CYP interaction (S13).

Q7 · Mechanism · Antacids, H2 Blockers & Proton Pump Inhibitors

Which antacid is most likely to cause diarrhea and accumulate in renal impairment?

- **A** Magnesium hydroxide
- **B** Famotidine
- **C** Calcium carbonate
- **D** Aluminum hydroxide
- **E** Sucralfate

ANSWER: A

Magnesium hydroxide can cause diarrhea; use caution in renal impairment because magnesium accumulates (S12; S54). S12; S54

- **B.** An H2 blocker, not an antacid (S13).
- **C.** Calcium carbonate constipates; milk-alkali with excess (S12).
- **D.** Aluminum is constipating (S12).
- **E.** A mucosal protectant; constipation (S20).

Q8 · Clinical vignette · Antacids, H2 Blockers & Proton Pump Inhibitors

A patient on clopidogrel after a stent needs long-term acid suppression. Which PPI does the textbook prefer?

- **A** Esomeprazole, the S-isomer of omeprazole
- **B** Cimetidine, since it is not a PPI
- **C** Omeprazole, the most studied prototype
- **D** Pantoprazole, with minimal CYP2C19 inhibition
- **E** Any PPI — none interact with clopidogrel

ANSWER: D

When PPIs are prescribed with clopidogrel, agents with minimal CYP2C19 inhibition (pantoprazole or rabeprazole) may be preferred (p. 1161). p. 1161

- **A.** Also inhibits CYP2C19 (S16).
- **B.** A broad CYP inhibitor (S13).
- **C.** Omeprazole inhibits CYP2C19 (S15).
- **E.** Several do (p. 1161).

Q9 · Mechanism · Mucosal Protectants & H. pylori Eradication

A patient needs chronic NSAIDs and has had an NSAID ulcer. How does misoprostol protect, and what is its major contraindication?

- **A** Blocks H2 receptors; avoid in older adults
- **B** Neutralizes acid; separate from other drugs
- **C** PGE1 analog restoring prostaglandin defense; never in pregnancy
- **D** Forms a barrier over the ulcer base; avoid in CKD
- **E** Blocks the proton pump; avoid with clopidogrel

ANSWER: C

Misoprostol, a PGE1 analog, restores the prostaglandin tone NSAIDs suppress; it is contraindicated in pregnancy (abortifacient) (S20; SG U3-R5). S20; SG U3-R5

- **A.** That is famotidine (S13).
- **B.** That is antacids (S12).
- **D.** That is sucralfate (S20).
- **E.** That is omeprazole (S15).

Q10 · Clinical vignette · Mucosal Protectants & H. pylori Eradication

A patient finishes H. pylori therapy and wants a test of cure next week while still on the PPI. What is the right advice?

- **A** Repeat the full regimen without testing
- **B** Test during therapy to adjust antibiotics
- **C** No testing is ever needed after therapy
- **D** Test now; the PPI improves test accuracy
- **E** Test ≥ 4 weeks after therapy, PPI held 2 weeks

ANSWER: E

Confirm eradication at least 4 weeks after therapy and off the PPI for 2 weeks — PPIs cause false negatives (S23; S24). S23; S24

- **A.** Test first (S23).
- **B.** Test after completion (S23).
- **C.** Confirm eradication (S23).
- **D.** PPIs cause false negatives (S23).

Q11 · Clinical vignette · Mucosal Protectants & H. pylori Eradication

A patient on simvastatin is about to start clarithromycin-based triple therapy. What is the concern?

- **A** CYP3A4 inhibition raises the statin level
- **B** Tooth discoloration in adults
- **C** Simvastatin inactivates clarithromycin
- **D** A disulfiram-like reaction with alcohol
- **E** The PPI binds simvastatin in the gut

ANSWER: A

Clarithromycin inhibits CYP3A4 and must not be combined with simvastatin, lovastatin, or ergotamine (S24; S60; S71). S24; S60

- **B.** Tetracycline in children (S62).
- **C.** The interaction runs the other way (S24).
- **D.** That is metronidazole (S24).
- **E.** Not the taught interaction (S24).

Q12 · Clinical vignette · Mucosal Protectants & H. pylori Eradication

A patient on metronidazole for H. pylori plans to drink wine at a wedding. What should they be told?

- **A** Stop the PPI on the day of drinking
- **B** Avoid alcohol — a disulfiram-like reaction can occur
- **C** Alcohol is fine; only tetracycline interacts
- **D** Take the dose with milk to prevent problems
- **E** Alcohol raises clarithromycin resistance

ANSWER: B

Metronidazole plus alcohol causes a disulfiram-like reaction; avoid alcohol during and for 3 days after (S24; S61). S24; S61

- **A.** Not the counseling point (S24).
- **C.** Metronidazole interacts with alcohol (S24).
- **D.** Milk chelates tetracycline, not this (S62).
- **E.** Not the taught risk (S24).

Q13 · Mechanism · Mucosal Protectants & H. pylori Eradication

Why must sucralfate be separated from PPIs and antacids?

- **A** It irreversibly blocks the proton pump itself
- **B** It is a PGE1 analog that needs alkaline pH
- **C** It needs an acidic environment to polymerize and bind
- **D** It neutralizes the antacid in the stomach
- **E** It inhibits CYP2C19, raising the PPI level

ANSWER: C

Sucralfate polymerizes in acid (pH below 4) to bind the ulcer base; give on an empty stomach, separated from antacids/PPIs (S20–S21). S20; S21

- **A.** That is the PPI mechanism (S15).
- **B.** Misoprostol is the PGE1 analog (S20).
- **D.** Not the taught reason (S20).
- **E.** Not a sucralfate effect (S20).

Q14 · Mechanism · Antiemetics

Which pairing correctly matches an antiemetic with its target?

- **A** Dronabinol — NK1 receptor
- **B** Ondansetron — D2 receptor
- **C** Metoclopramide — H1 receptor
- **D** Aprepitant — NK1 receptor
- **E** Meclizine — 5-HT3 receptor

ANSWER: D

Aprepitant blocks NK1; ondansetron 5-HT3; meclizine H1/M1; metoclopramide D2 (S30; SG U3-M). S30; SG U3-M

- **A.** Dronabinol activates CB1 (S38).
- **B.** Ondansetron blocks 5-HT3 (S30).
- **C.** Metoclopramide blocks D2 (S30).
- **E.** Meclizine blocks H1/M1 (S30).

Q15 · Clinical vignette · Antiemetics

A patient with diabetic gastroparesis has taken metoclopramide for 6 months and now has involuntary lip-smacking movements. What explains this?

- **A** Acute dystonia that resolves with time
- **B** CB1 activation causing dysphoria
- **C** Tardive dyskinesia (boxed warning)
- **D** Serotonin syndrome from 5-HT3 blockade
- **E** Hypomagnesemia from long-term use

ANSWER: C

Metoclopramide carries a boxed warning for tardive dyskinesia with prolonged use or high doses; avoid beyond 12 weeks (S36–S37; SG U3-R8). S36; S37; SG U3-R8

- **A.** Acute dystonia occurs early; this is late and long-term (S36).
- **B.** A cannabinoid effect (S38).
- **D.** Ondansetron's interaction (S31).
- **E.** A PPI risk (S15).

Q16 · Clinical vignette · Antiemetics

A patient on citalopram and with a borderline long QT needs an antiemetic after surgery. Why is ondansetron a cautious choice?

- **A** Uterine contractions
- **B** Tardive dyskinesia after one dose
- **C** Disulfiram-like reaction with alcohol
- **D** QT prolongation and serotonin syndrome risk
- **E** Rebound acid hypersecretion

ANSWER: D

Ondansetron prolongs the QT interval and carries serotonin syndrome risk with serotonergic drugs (S31). S31

- **A.** Misoprostol (S20).
- **B.** A long-term metoclopramide risk (S36).
- **C.** Metronidazole (S24).
- **E.** Stopping PPIs (S15).

Q17 · Mechanism · Antiemetics

A patient with refractory chemotherapy nausea is started on dronabinol. How does it work?

- **A** Activates CB1 receptors in the vomiting center and CTZ
- **B** Blocks substance P at NK1 receptors
- **C** Blocks muscarinic receptors in the vestibular system
- **D** Blocks 5-HT3 on vagal afferents
- **E** Blocks D2 in the chemoreceptor trigger zone

ANSWER: A

Dronabinol is a CB1 agonist that reduces neurotransmitter release and stimulates appetite — distinct from 5-HT3, D2, NK1, H1/M1 (S38–S39; SG U3-M). S38; S39; SG U3-M

- **B.** Aprepitant (S30).
- **C.** Scopolamine (S30).
- **D.** Ondansetron (S31).
- **E.** Metoclopramide (S36).

Q18 · Mechanism · Antiemetics

Why are meclizine and scopolamine the classes that fit motion sickness?

- **A** They activate CB1 receptors in the cortex
- **B** They block D2 receptors in the CTZ
- **C** They block 5-HT3 on gut vagal afferents
- **D** They block NK1 receptors in the CTZ
- **E** They block H1/M1 input from the vestibular system

ANSWER: E

Motion sickness is vestibular; H1/M1 antagonists block histamine/acetylcholine from the vestibular system (S30; S33 notes; SG U3-R7). S30; S33; SG U3-R7

- **A.** That is dronabinol (S38).
- **B.** That is metoclopramide/prochlorperazine (S30).
- **C.** That is ondansetron (S30).
- **D.** That is aprepitant (S30).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO3.1** Identify major pharmacologic classes used in GI and nutrition conditions and describe their mechanisms of action.
- [ ] **IO3.2** Describe the indications, contraindications, adverse effects, and clinically significant interactions of GI and nutrition medications.
- [ ] **IO3.3** Compare agents within and across these classes to distinguish clinically relevant differences in efficacy, safety, and use.
- [ ] **IO3.4** Select appropriate pharmacologic therapy for common GI and nutrition conditions based on clinical presentation and patient-specific factors.
- [ ] **IO3.5** Identify appropriate dosing, administration, and monitoring considerations for GI and nutrition medications.
- [ ] **IO3.6** Explain important medication-use and safety information to patients receiving GI and nutrition pharmacotherapy.

- [ ] **SO1** Identify the major pharmacologic classes used to manage acid-peptic disease and nausea/vomiting, and describe their mechanisms of action.
- [ ] **SO2** Describe the indications, contraindications, adverse effects, and clinically significant interactions of antisecretory and mucosal-protective agents.
- [ ] **SO3** Compare agents within and across these classes to distinguish clinically relevant differences in efficacy and safety.
- [ ] **SO4** Select appropriate therapy for acid-peptic disease and nausea/vomiting based on clinical presentation and patient-specific factors.
- [ ] **SO5** Identify dosing, administration, and monitoring considerations, and explain key medication-safety counseling points to patients.
- [ ] **SO6** Preview: GI motility disorders, IBD pharmacotherapy, and Nutrition/Dietary Supplements continue in Session 2 (Thursday, November 5).